IRP Employability Intelligence Platform
Developer Handover Document (MVP v1.0)

Executive Summary

This document is the MVP handover spec for the Employability Intelligence Platform (IRP): a data-driven system to measure, improve and prove student employability through structured assessments, cohorting and tracked interventions. The pilot covers one college and ~500 students. The MVP builds authentication, student/college management, an assessment and scoring engine, rule-based cohort assignment, an intervention library with session tracking, and reporting dashboards — explicitly excluding AI chat, mobile apps, gamification and predictive analytics. Data flows through a PostgreSQL backend (Python/FastAPI, React frontend), with configurable placement-readiness thresholds and full audit logging. The first workflow to prove out is bulk upload → login → baseline assessment → scoring → cohort assignment → dashboard update. Success is defined as 500 students onboarded, 95% assessment completion, 85% intervention-tracking accuracy, 100% PDF report generation, and a 30-point improvement in placement readiness — all built on the founding principle of measuring and proving what interventions actually work.

1. Product Vision

Build a data-driven Employability Intelligence Platform that measures, improves, and proves student employability outcomes through structured interventions and analytics.

2. Pilot Scope

Pilot with one college and approximately 500 students. Assessments will be conducted in supervised windows and the system will track interventions and placement readiness.

3. Business Objectives

Measure baseline employability, segment students into cohorts, deliver and track interventions, measure improvement over time, provide placement readiness metrics, and identify interventions that work best.

4. MVP Modules (Build Only)

Authentication

College & Department Management

Student Management

Bulk Student Upload

Assessment Engine

Scoring Engine

Cohort Assignment Engine

Intervention Library

Intervention Session Tracking

Student Participation Tracking

Reports & Dashboards

5. Explicitly Out of Scope

AI Chat, Resume Analyzer, Mock Interview Engine, Mobile App, Gamification, Notifications, Recommendation Engine, Predictive Analytics.

6. User Roles

Super Admin (IRP): Manage colleges, intervention library, thresholds and dashboards.

College Admin: Upload students, create assessment windows, schedule interventions and view dashboards.

Trainer: Conduct intervention sessions, mark attendance and participation.

Student: Login, take assessments and view reports and progress.

7. Student Journey

Student Login → Baseline Assessment → Scores Generated → Cohort Assignment → Participate in Interventions → Mid Assessment → Final Assessment → View Progress.

8. Assessment Requirements

Assessment Types: Baseline, Mid and Final.

Requirements: Controlled windows, desktop/laptop only during pilot, no edits after submission, timestamps and submission logs.

9. Cohort Assignment

Placement Ready, Communication Deficit, Interview & Confidence Deficit and High Risk. Use rule-based assignment only in MVP.

10. Placement Readiness Rules

Store thresholds as configurable data and not hardcoded values.

Current hypothesis:

Communication ≥ 70

Interview Readiness ≥ 75

Professionalism ≥ 75

Overall Score ≥ 75.

11. Dashboards

Student Dashboard: Scores, cohort, improvement trends and placement readiness category.

College Dashboard: Placement readiness %, near ready %, high risk %, department scores and intervention participation.

Intervention Dashboard: Top interventions, lowest performing interventions, participation rates and average improvements.

12. Database Tables

Colleges, Departments, Students, AssessmentCycles, StudentAssessments, Cohorts, StudentCohorts, Interventions, InterventionSessions, StudentParticipation, StudentImprovement, InterventionEffectiveness, PlacementReadinessRules and AuditLog.

13. Core Database Philosophy

Every field must answer one of four questions:

1. Who is this student?

2. What intervention was delivered?

3. Did it work?

4. For whom does it work best?

Never overwrite history. All major activities must be time-stamped and auditable.

14. API Requirements

Authentication APIs

Student APIs

Assessment APIs

Cohort APIs

Intervention APIs

Dashboard APIs

PDF Report APIs

15. Non-Functional Requirements

JWT Authentication, PostgreSQL database, support for at least 500 concurrent students during assessment windows, audit logging, backup strategy and ability to scale to 10,000 students without redesign.

16. Technology Stack

Backend: Python + FastAPI

Database: PostgreSQL

Frontend: React

Hosting: Cloud VM initially

Authentication: JWT Tokens

Version Control: Git/GitHub

17. Expected Deliverables from Developer

Database ER Diagram

API Documentation

UI Wireframes

Development Plan

Sprint Plan

Deployment Architecture

Source Code Repository

Testing Plan

User Acceptance Testing Plan

18. First Workflow to Build

Bulk Upload Students → Student Login → Baseline Assessment → Score Calculation → Cohort Assignment → Dashboard Update.

No additional features should be built until this flow works flawlessly.

19. Success Criteria for Pilot

500 students onboarded.

95% assessment completion.

85% intervention tracking accuracy.

100% PDF report generation.

Dashboard actively used by management.

Target placement readiness improvement: 30 percentage points.

20. Development Approach

Work in weekly sprints with demonstrations every week. Focus on clean data capture and reliability before adding advanced capabilities.

Founding Principles (Non-Negotiable)

• Measure. Improve. Prove.

• Data quality is more important than UI aesthetics.

• Interventions are experiments and must be measurable.

• The software should learn what interventions work, for whom and by how much.

• No feature should be added unless it improves data quality, intervention quality, outcome measurement or decision making.